

Ian Pang

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EDUCATION

Purdue University – West Lafayette

West Lafayette, IN

Bachelor of Science in Computer Science

Expected May 2028

- Relevant Coursework: Object Oriented Programming (Java), Discrete Math, Programming in C, Computer Architecture, Data Structures and Algorithms, Systems Programming, Introduction To AI (AI/ML)

EXPERIENCE

Incoming Software Engineer Intern

Sept 2026 – Nov 2026

Amazon Web Services (AWS)

Boston, MA

- Incoming Software Engineer Intern on the Elastic Block Store (EBS) team.

Software Engineer Intern

June 2026 – Present

American Family Insurance

Boston, MA

- Designed an AWS pipeline (Python, Lambda, API Gateway, S3, DynamoDB) to recover failed records in a agent-onboarding system, replacing manual fixes with a single API call, keeping PII data within AWS.
- Owned the system end to end with a full-stack application (Next.js, REST API) for internal users to self-serve record reprocessing and real-time agent-status checks along with Entra ID, OAuth 2.0 SSO authentication.
- Architected CI/CD deployment pipelines and modular Infrastructure as Code templates to deploy the entire system (Fargate, ECS Cluster, Route53, Docker) for internal use, routing traffic through private ALBs.
- Standardized Datadog observability across 20+ corporate gateway endpoints for 6 insurance partners, implementing custom latency metrics and time-series dashboards to detect bottlenecks.

Research Assistant

May 2025 – Present

Zhan Lab, Purdue University

West Lafayette, IN

- Engineered a real-time embryo detection pipeline using OpenCV and PyTorch to process 120 FPS high-resolution microfluidic video, achieving 20 ms end-to-end latency from detection to inference.
- Trained a ResNet-based CNN to classify embryo developmental stages with 80% accuracy based on 1,000+ embryo samples, and integrated Arduino control to direct viable embryos to separate channels for experimentation.

Software Engineer Intern

June 2025 – Aug. 2025

Suga International Holdings Limited

Hong Kong SAR

- Developed core digital signal processing features in C for an audio mixer, reducing CPU utilization by 40%.
- Minimized system storage footprint by 70% with compression techniques, achieving sub-2MB deployment.
- Designed and launched a dual-core MicroPython architecture with a custom WebSocket-based control server, enabling real-time audio command execution and decreasing operational latency from 120ms to 13ms.
- Implemented CI/CD build and test pipelines accelerating development iterations across the product lifecycle.

PROJECTS

Nadir | (1st Place HackIllinois) *Solana, TypeScript, Python, Kubernetes, S3 Object Storage*

- Engineered a dynamic tile server using TiTiler to perform on-the-fly COG processing of satellite imagery, enabling real-time, low-latency distribution of geospatial data to Mapbox web clients.
- Architected a scalable, cloud-native infrastructure using DigitalOcean Kubernetes to orchestrate PyTorch ML microservices for high availability and seamless auto-scaling for traffic.
- Optimized geospatial data pipelines by leveraging DigitalOcean Spaces (S3) for high-throughput storage, reducing data retrieval latency for large-scale structural damage detection and analysis.

Canopy | (2nd Place TartanHacks at CMU) *XRP Ledger, LangGraph, Python, FastAPI, Next.js, Supabase*

- Engineered an AI agent using LangGraph and an ML model to monitor live weather and satellite data, achieving 85% confidence thresholds for claim-less payouts.
- Developed escrow logic and NFT minting on the XRP Ledger to secure insurance funds and issue digital proof-of-coverage, allowing for instant payments.
- Built a full-stack architecture and a FastAPI backend with PostgreSQL for storing user data and agent reasoning.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, R, HTML, CSS, SQL/NoSQL

Frameworks: React, Node.js, Flask, FastAPI, PyTorch, REST APIs, OAuth, WebSockets

Tools: AWS, GCP, Docker, Kubernetes, CI/CD Pipelines, Git, PostgreSQL, PyTorch, OpenCV, LangChain

Miscellaneous: Distributed Systems, System Design, Backend, Cloud Infrastructure, Database, AI/ML, Agile Methodologies, Networking, Data Management

Hackathon Wins: TartanHacks (Canopy), HackIllinois (Nadir), HackUMass